

Title: Particles on the move

Starter

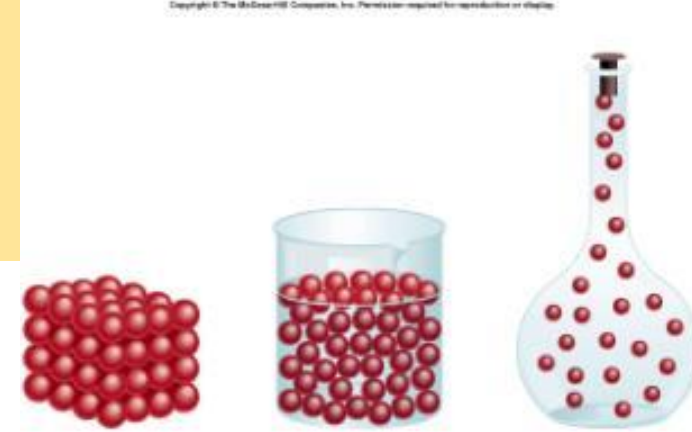
1. Draw the particle diagram for solids, liquids and gas

1. Which particles are able to move?

1. And how do they move?

Particles on the move

Starter



1. Draw the particle diagram for solids, liquids and gas

1. Which particles are able to move? **Gas**

1. And how do they move? **Gas particles move randomly**

Success criteria

Lesson Title	Working towards	Working at	Working beyond
Recap: Forces & Motion (Y7)	I can identify different forces acting on an object.	I can describe what balanced and unbalanced forces are.	I can explain how unbalanced forces change an object's motion using force diagrams.
Speed	I can identify the correct units for speed, distance, and time.	I can use the speed = distance $\div$ time formula to calculate speed, distance or time.	I can solve speed questions using different units and rearranged formulas.
Describing Movement	I can read values from a distance-time graph.	I can describe motion based on the shape of the graph.	I can draw and explain distance-time graphs from written data or scenarios.
Turning Forces	I can give examples of things that turn when a force is applied.	I can use the moment formula to calculate turning forces.	I can explain how balanced and unbalanced moments affect turning.
Pressure between solids	I can state that heavier or larger objects create more pressure.	I can calculate pressure using the correct formula and units.	I can compare pressure in different situations using scientific reasoning.
Pressure in liquids and gases	I can describe how pressure changes with depth in liquids and altitude in gases.	I can explain what causes pressure in liquids and gases using particles.	I can calculate liquid pressure using the $h\rho g$ formula and explain examples.
Particles on the move	I can define diffusion.	I can describe how temperature and concentration affect diffusion.	I can explain diffusion using the particle model and apply it to real examples.

States of matter

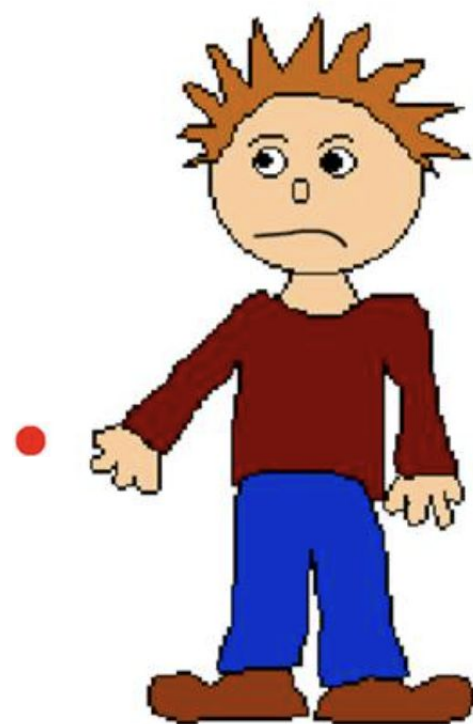
- **Gas** particles move freely, quickly and randomly (what does this mean?)
- **Liquid** particles move freely by sliding past each other
- **Solid** particles cannot move freely, they only vibrate in a fixed position

Write this down in your book!

What happens when the hamburger is taken out of the box? HOW do we smell it?

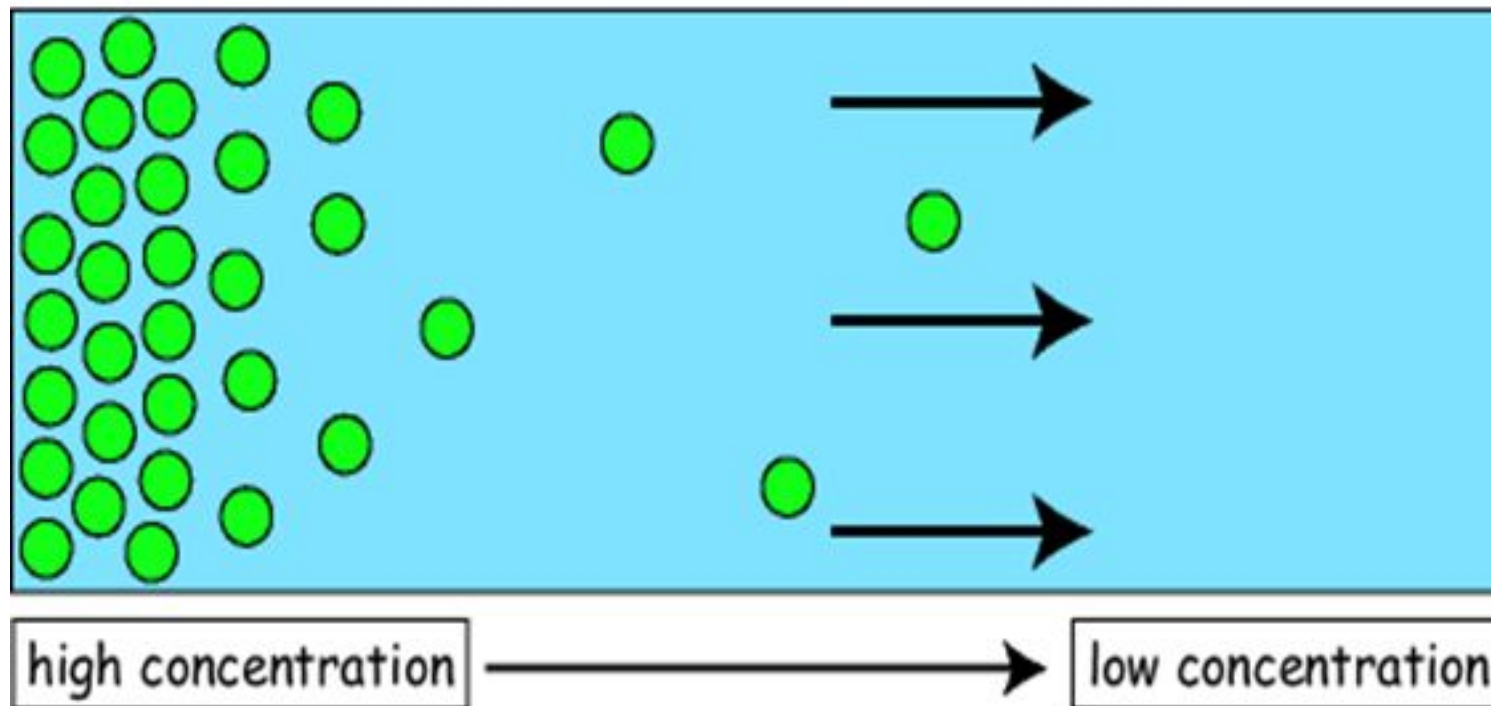






How we smell things

- Gas particles move and spread out (randomly), and they fill all the space available....
- When smelly ones reach your nose you can smell them :-)



Diffusion

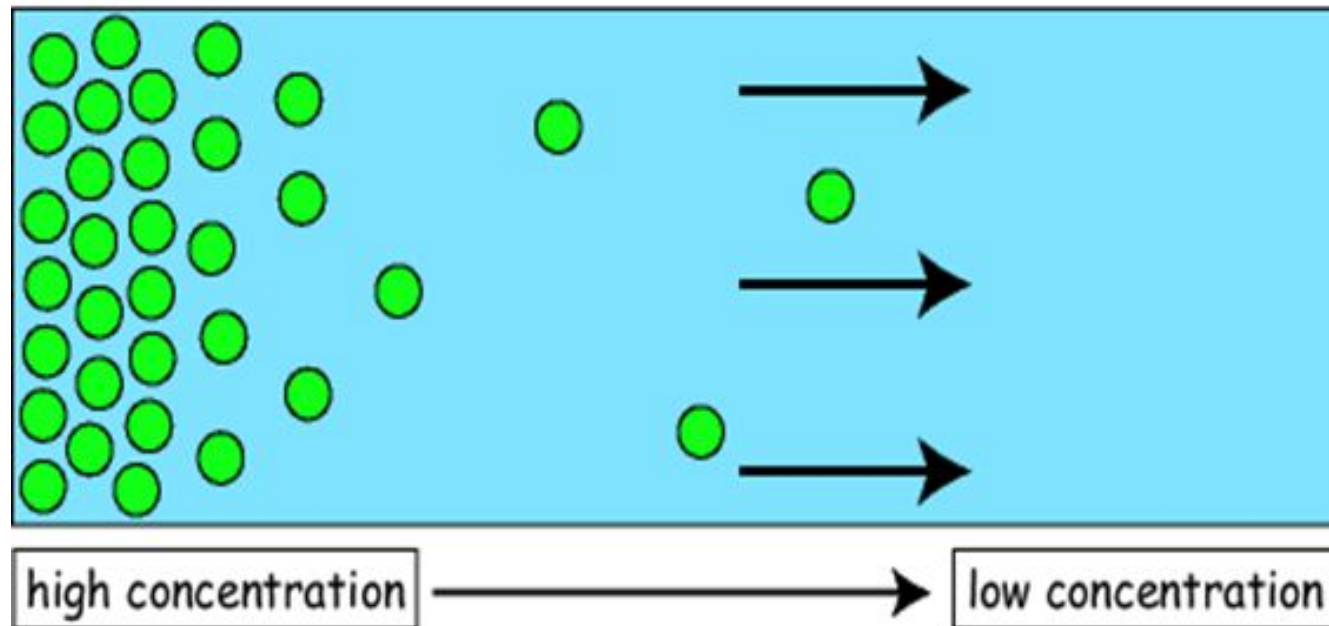
- How would you describe what diffusion is?



Diffusion

Write this down in your book!

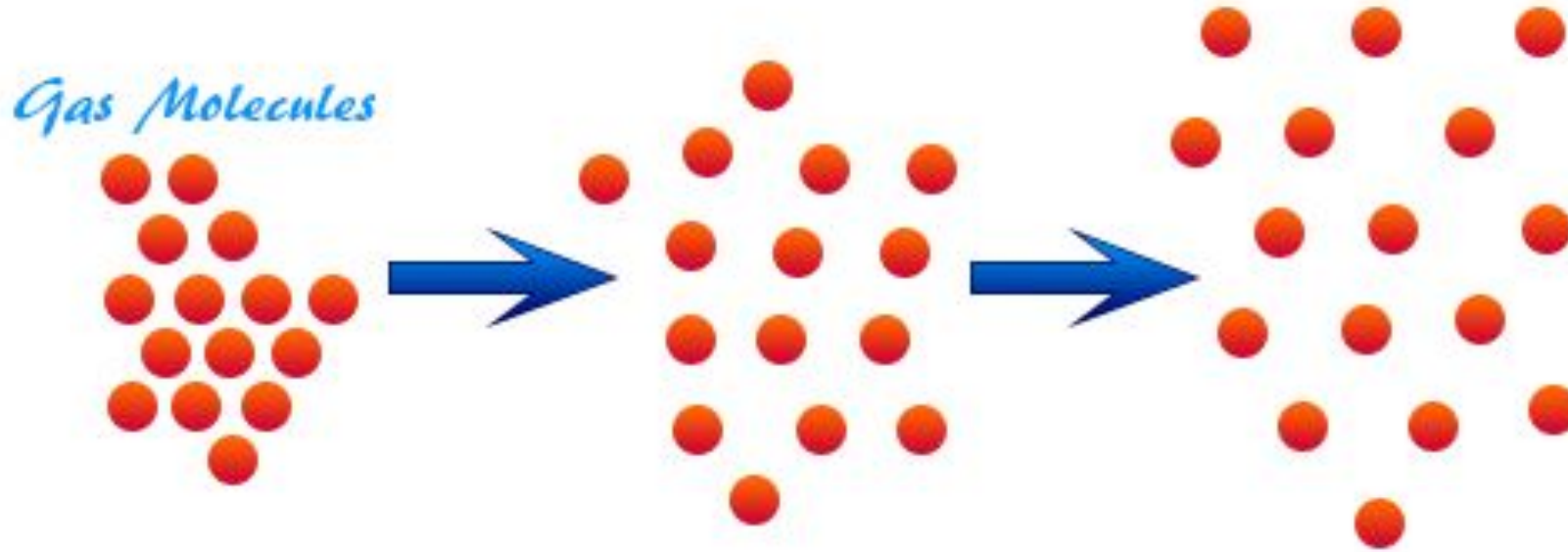
The movement of particles from a region of **higher concentration** to a region of **lower concentration** as a result of their **random** movement down a **concentration gradient**.



Concentration

Write this down in your book!

- When there are lots of particles together we say it is concentrated

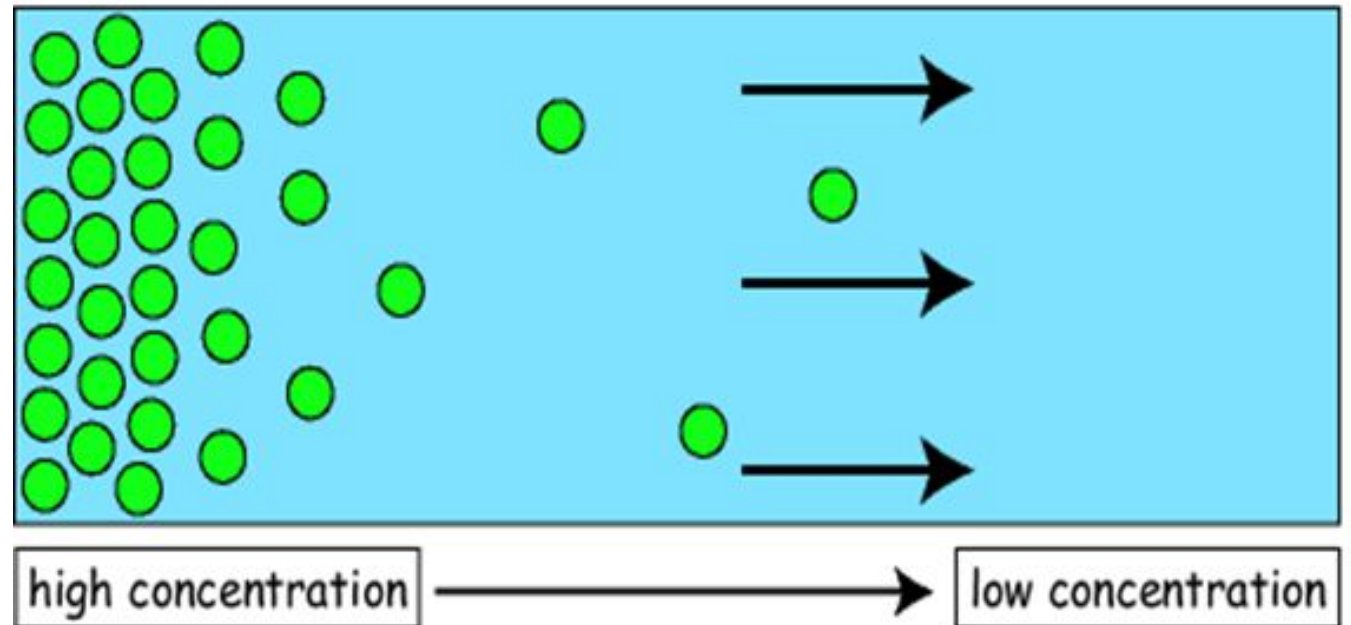


Where is the high concentration and where is the low concentration?

Diffusion

Talk to the person next to you...

1. Does diffusion 'stop'? If so, when?
2. Do the particles stop moving?



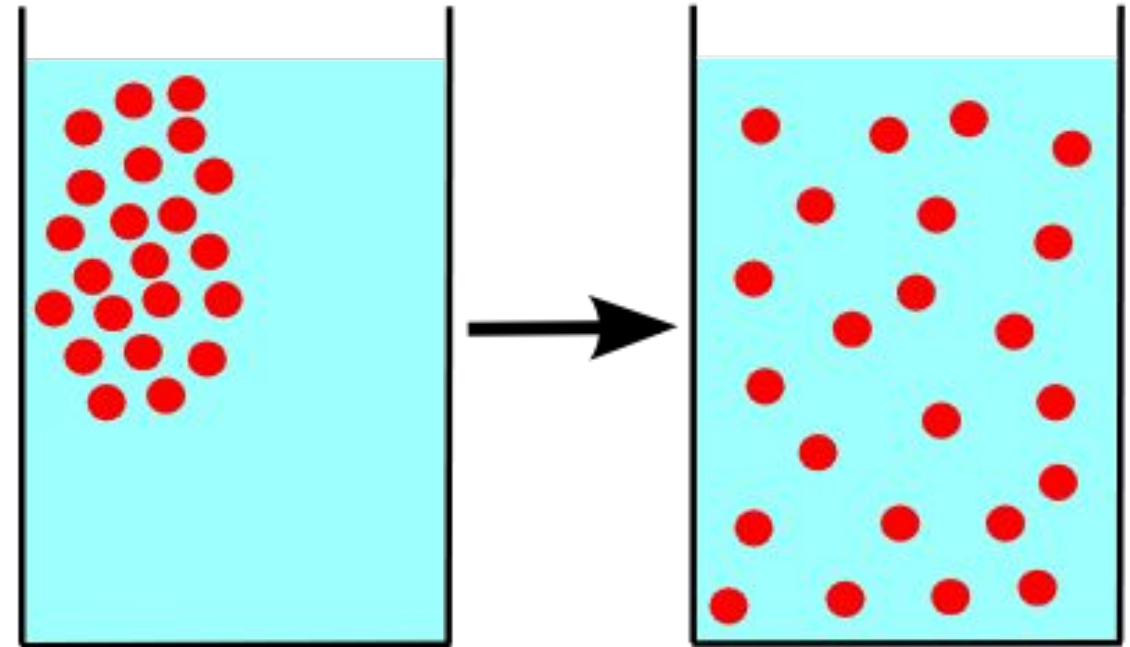
Diffusion

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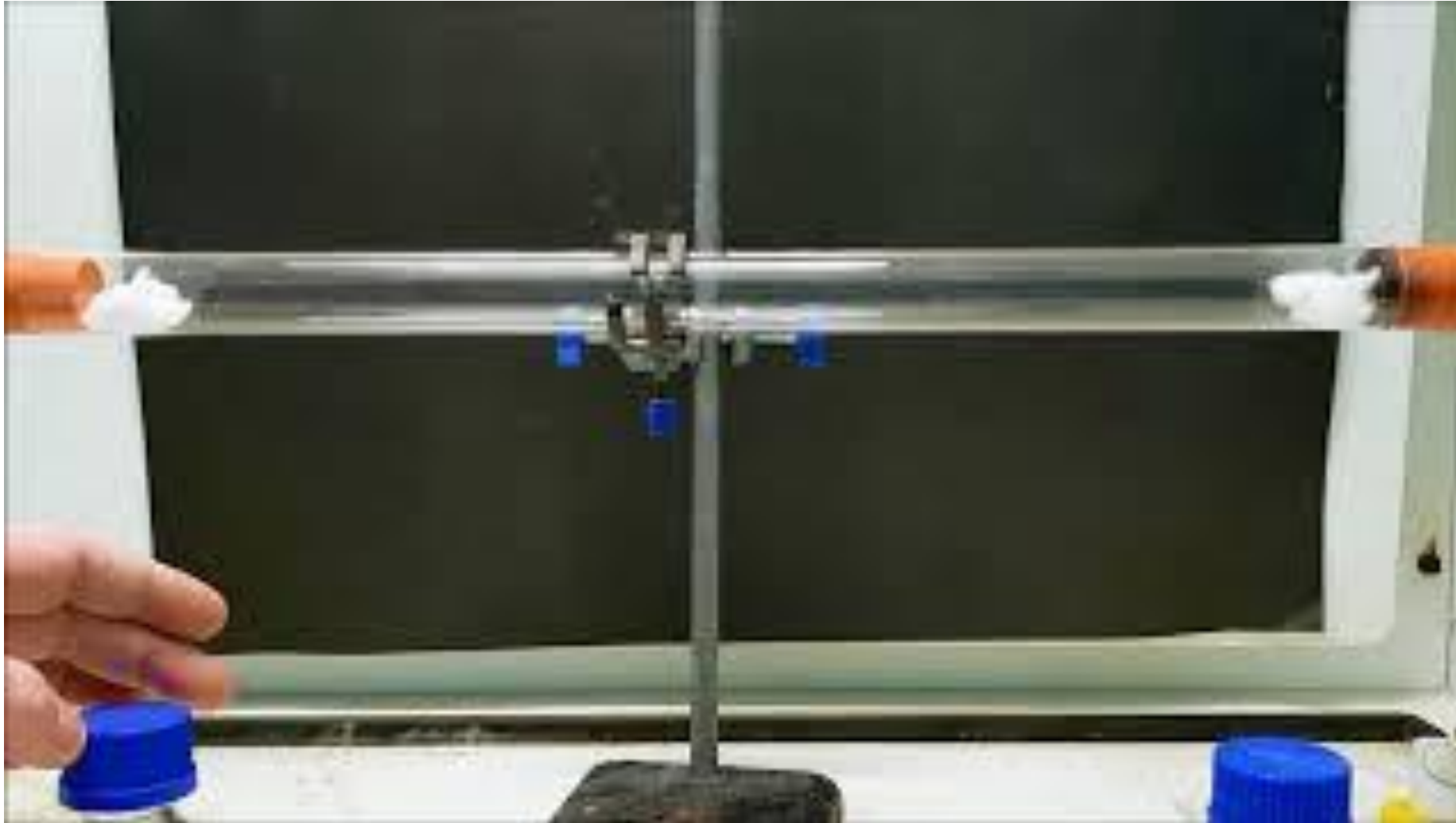
1. Does diffusion 'stop'? If so, when?
2. Do the particles stop moving?

**BUT the particles
never stop moving**

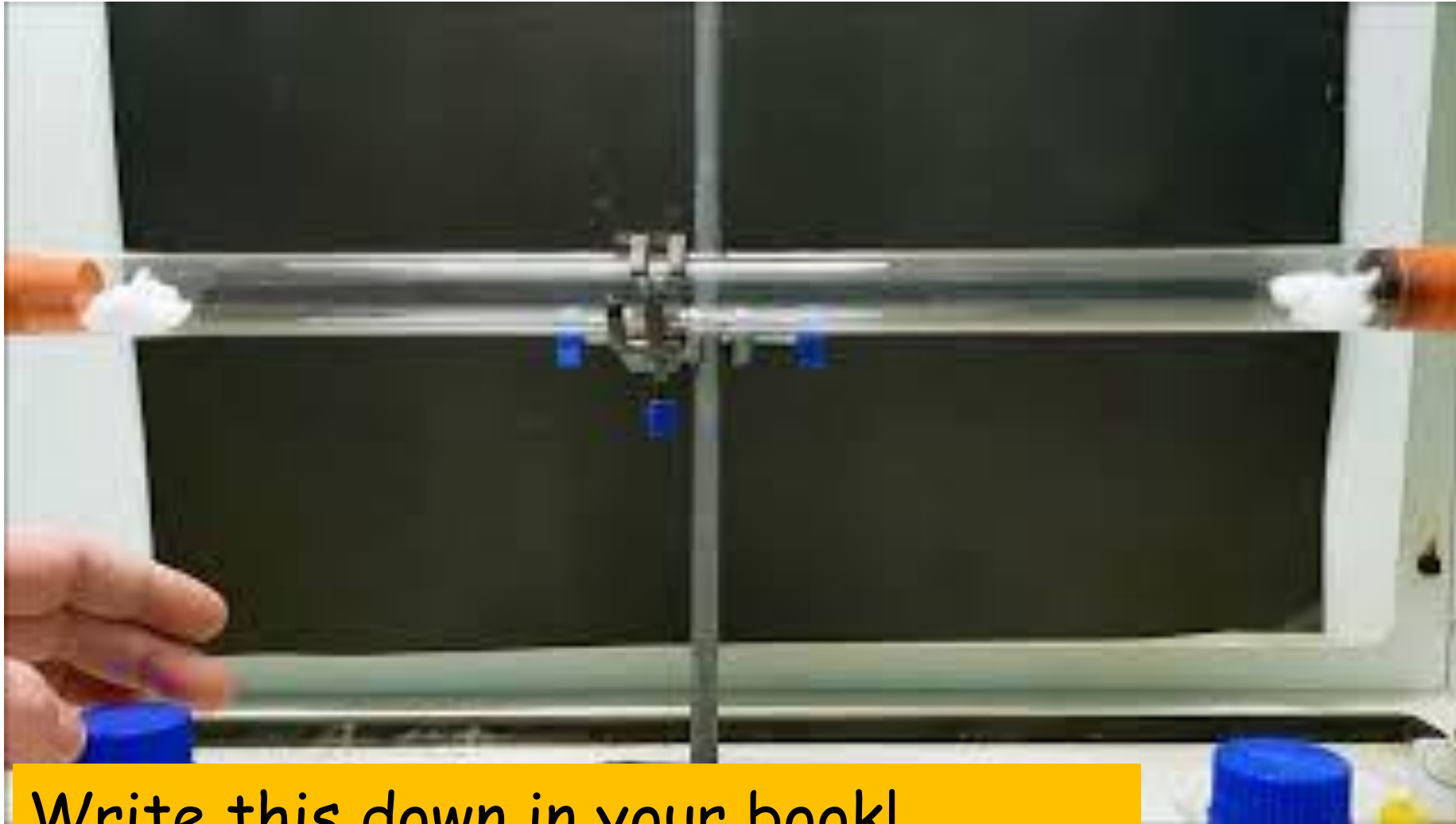
**When the particles are
EVENLY spread out, there will
be no OVERALL movement**



Watch this video on Ammonia and Hydrogen chloride



Demo 1 on Ammonia and Hydrogen chloride



Ammonia is lighter in mass than hydrogen chloride (HCl) therefore diffuse faster than hydrogen chloride (HCl) forming a white ring closer to the HCl side.

Write this down in your book!

What do we know about diffusion

- Diffusion happens in _____s and _____s.
- Diffusion cannot happen in _____s because the particles cannot _____.
- Diffusion happens quicker in _____ because...

Write this down in your book!

What do we know about diffusion

- Diffusion happens in **gases** and **liquids**.
- Diffusion cannot happen in **solids** because the particles cannot **move**.
- Diffusion happens quicker in **gases** because **the particles in gases are able to move more freely than liquids**

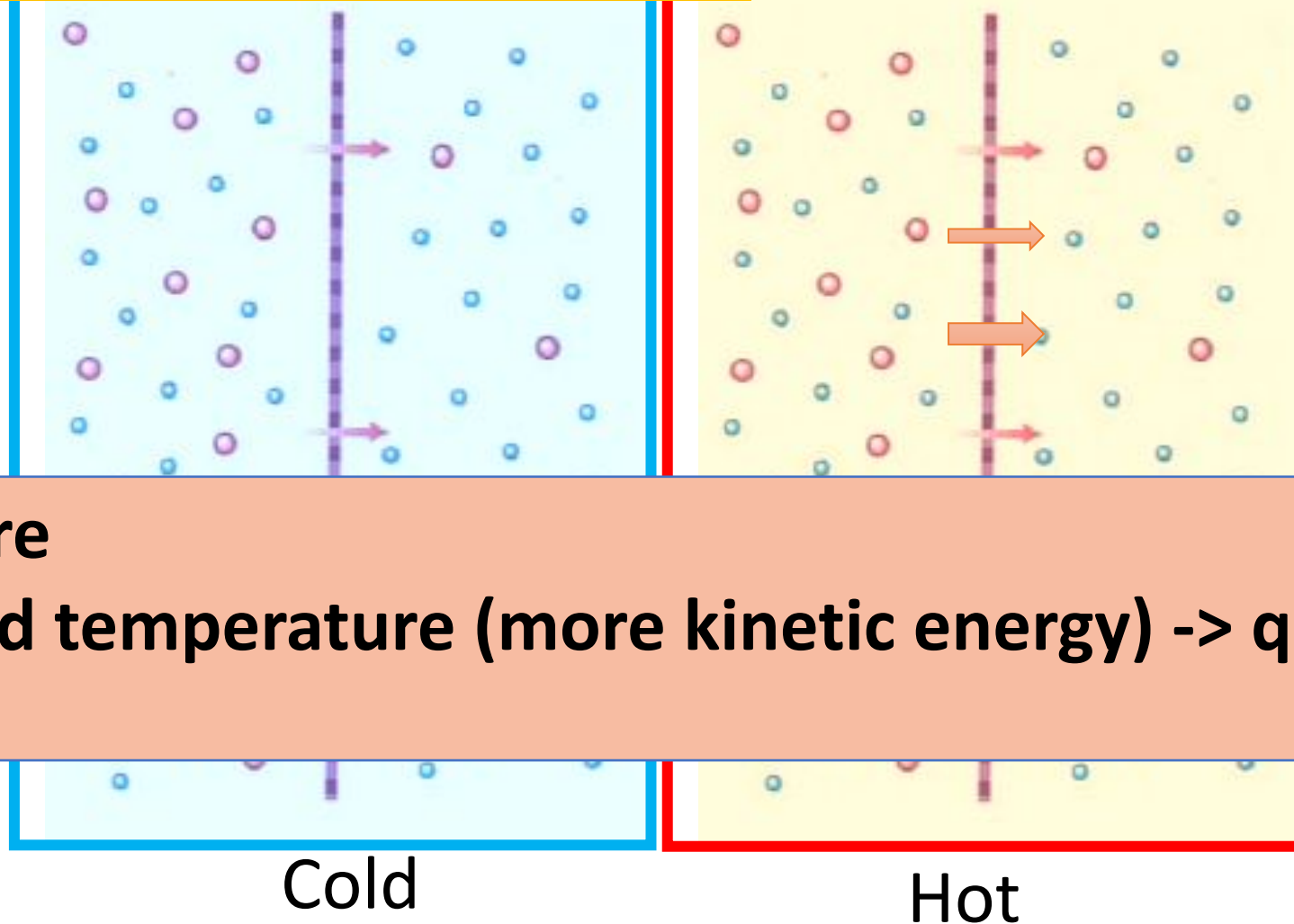
Measuring the Speed of Diffusion

What do you think might affect how quickly diffusion happens?

How could we measure the speed of diffusion?

Factors which affect the rate of diffusion

Write this down in your book!

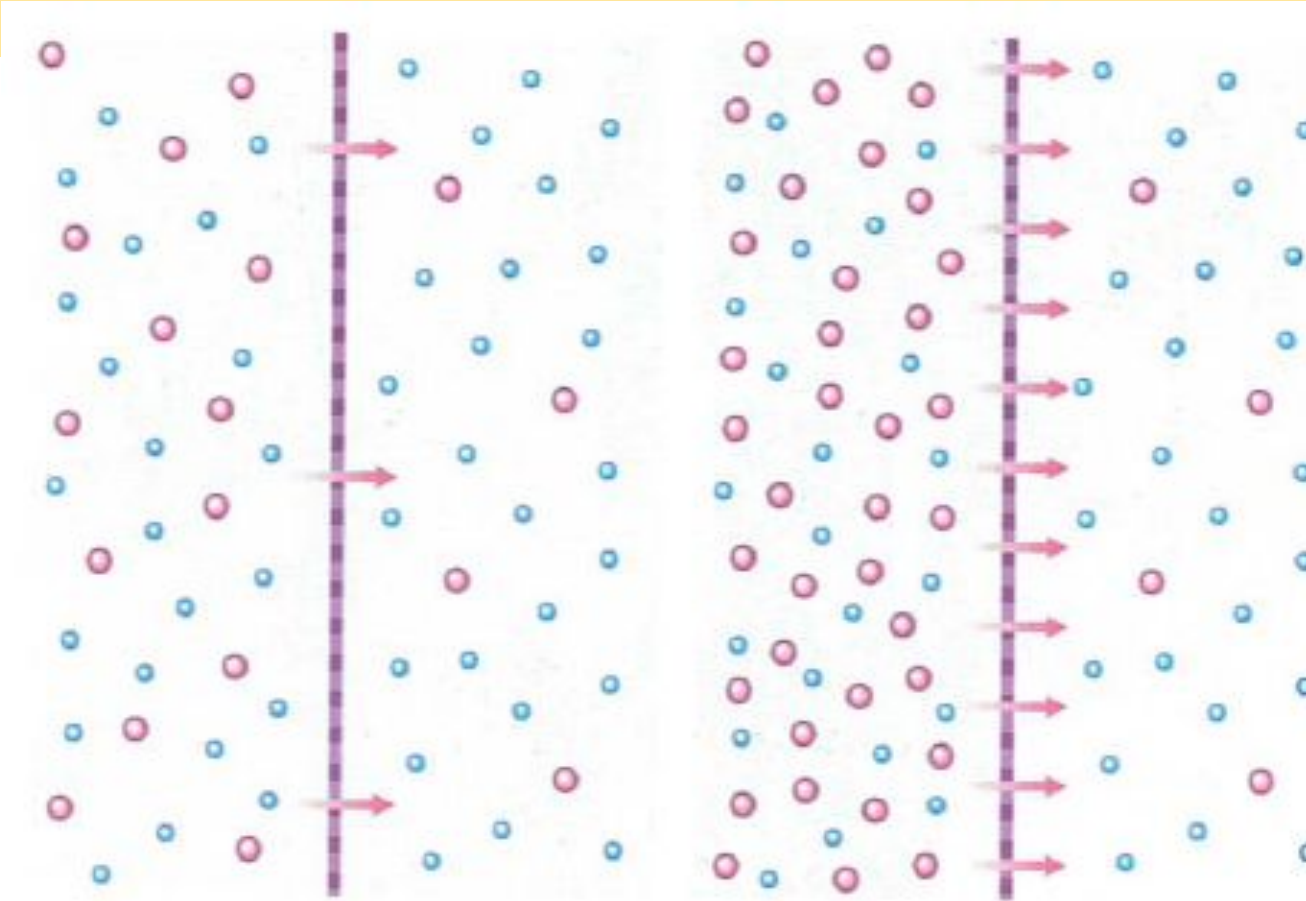


Temperature

Increased temperature (more kinetic energy) -> quicker rate

What else affects the rate of diffusion?

Write this
down in your
book!



The bigger the
difference in
concentration
the faster the
diffusion
happens.

Concentrations gradient

Increased gradient -> quicker rate

Questions

- 1 Zara's mother opens a bottle of perfume. Zara is at the other side of the room.

After a few minutes, Zara can smell the perfume.

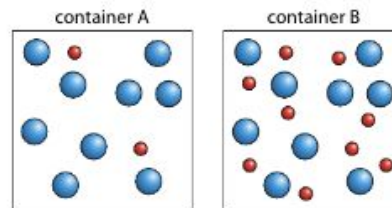
- a Which of these explains why Zara smells the perfume?
Write **one** letter.
- A All the gas particles of the perfume move in one direction.
 - B All the gas particles of the perfume move randomly.
 - C All the particles in the air move in one direction.
 - D All the particles in the air are stationary (not moving).
- b Which of these changes would result in Zara smelling the perfume in a shorter time?

In each case, **no** other variable changes.

Write the letters for **all** the correct statements.

- A The perfume in the bottle is at a higher temperature.
 - B Zara moves further away from the bottle of perfume.
 - C Zara's mother puts the lid on the perfume bottle.
 - D The air in the room is at a lower temperature.
- 2 Describe what is meant by the term 'diffusion'.
- 3 The diagram shows two types of particle in containers, A and B. The containers have the same volume.

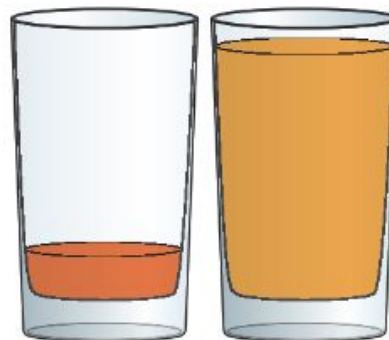
- a Explain whether the particles in the containers are in a solid, liquid or gas state.
- b Explain how the concentrations of the blue particles compare in containers A and B.
- c Explain how the concentrations of the red particles compare in containers A and B.



- 4 Sofia is making an orange flavour drink. She pours a small volume of concentrated orange juice into a glass. She then adds water until the glass is full.

- a Explain how the orange colour from the juice spreads into the water.
- b Sofia says: 'When the orange colour has stopped spreading, the particles in the liquid have stopped moving.'

Explain whether Sofia is correct.



- 5 Which of these will result in diffusion?

Write the letters for **all** the correct statements.

- A Adding milk to coffee.
- B Adding sand to water.
- C Adding salty water to pure water.
- D Allowing gas from a cylinder to escape into the air.
- E Throwing small pieces of rock into the air.

Learner's book pg 117

Complete this in your learner's book or in your notebook!

Markscheme will be posted on google classroom at 2.40pm

soft copy of the learner's book is on the next few slides

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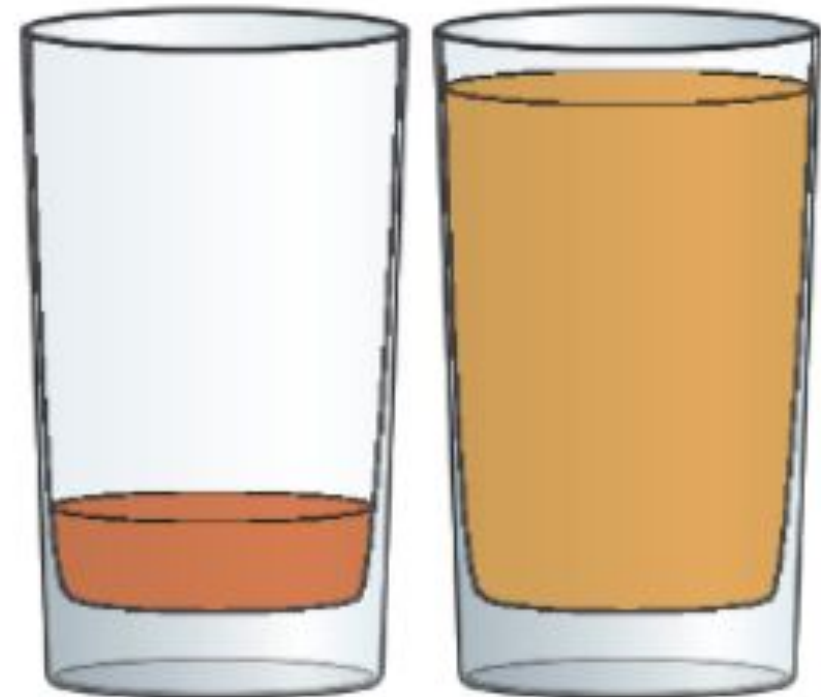
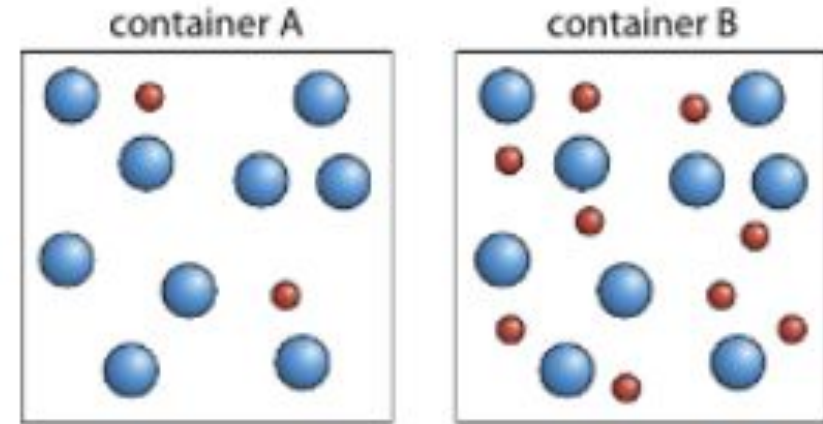
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-

Extra time? Let's practice...

Workbook page 64-68

Markscheme will be posted at
2.50pm on google classroom

mark your work using a green pen.

Extension task:
complete workbook page 47-51

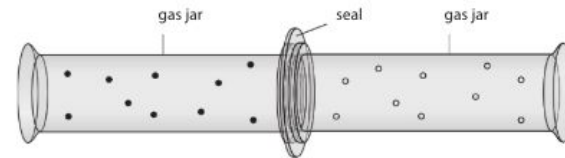
> 3.7 Particles on the move

Exercise 3.7A Diffusion in gases and liquids

Focus

In this exercise, you will describe what happens when gases mix and when liquids mix.

- 1 Two glass jars are kept apart with a seal.
One jar contains a coloured gas.
The other jar contains a colourless gas.



The seal is removed.

Draw the arrangement of particles in the jars some time after the seal was removed.

